MGU-UGP (HONOURS)REGULAR EXAMINATION

OCTOBER 2025

THIRD SEMESTER

INTRODUCTION TO PROGRAMMING

SCHEME

Part A – Short Answer Type Questions

1. **Describe I/O operator.**

I/O operators in C++ are:
<< (insertion operator): used with cout to output data.
Example: cout << "Hello";
>> (extraction operator): used with cin to input data.
Example: cin >> x;
These operators are overloaded in the iostream library.

2. **Estimate the advantage of using a structured programming language.**

Easier to read, debug, and maintain.
Programs are modular (divided into functions).
Reduces code redundancy.
Promotes top-down design and logical program flow.

3. **Interpret the meaning of polymorphism.**

Polymorphism means “many forms”.
It allows the same function or operator to behave differently based on the context.
Types:
Compile-time (function overloading, operator overloading)
Run-time (function overriding using virtual functions)

4. **Present the difference between Break and Continue in loops.**

Break
Terminates the entire loop immediately.
Used when an exit from the loop is required.
Example: exit on condition
Continue
Skips the current iteration and continues with the next.
Used to skip specific conditions inside loop.
Example: skip negative values

5. **Describe how you can recognize that a function belongs to a class.**

A function that belongs to a class is called a member function.
It is declared inside the class definition.
Accessed using the dot (.) operator with an object.
Example: object.functionName();

6. **Examine how 2-D array can be represented in memory.**

Stored in row-major order (row by row in continuous memory).
Example:
int a[2][3] = {{1,2,3},{4,5,6}};
Memory representation:
a[0][0], a[0][1], a[0][2], a[1][0], a[1][1], a[1][2]
Address formula:
Address(a[i][j]) = Base + ((i * totalColumns) + j) * size_of_element

7. Compare friend function and member function.

Friend Function
Declared using friend keyword.
Not a member of the class but can access private data.
Invoked like a normal function.
Member Function
Declared normally inside class.
Is a member and can access all members directly.
Invoked using object and dot operator.

8. Differentiate between member function and non-member function.

Member Function
Declared inside the class.
Accesses class data directly.
Called using object.
Non-Member Function
Declared outside the class.
Cannot access private data (unless friend).
Called independently.

9. Mention two characteristics of Constructor.

1. Constructor has same name as the class.
2. It is automatically called when an object is created.
3. It has no return type (not even void).
4. Used to initialize data members.
(Any two points fetch full marks.)

10. Define parameterised constructor.

A parameterised constructor is a constructor that takes arguments to initialize object data members.
Example:

```
class Student {
   int age;
public:
   Student(int a) { age = a; }
};
```

Allows different initial values for different objects.

11.Why is virtual base class used?

To avoid ambiguity in multiple inheritance (diamond problem). Ensures only one copy of base class is inherited by all derived classes.
Example:
class A{}; class B: virtual public A{}; class C: virtual public A{}; class D: public B, public C{};

12.Write any two rules of operator overloading.

- Only existing operators can be overloaded.
- At least one operand must be a user-defined type.
- Precedence/associativity of operators cannot change.
- Some operators like '.', '?:', 'sizeof' cannot be overloaded.

Part B – Short Essay Type Questions

13. Symbols used in Flowchart:

Oval – Start/End
Parallelogram –
Input/Output Rectangle –
Process Diamond –
Decision
Arrow – Flow of control
Connector – Joins flow lines

14. Differentiate between Testing and Debugging:

Testing – Finding errors, done after coding.
Debugging – Correcting errors, done by programmers. Testing validates; debugging corrects.

15. Can access specifiers be used more than once in a class?
Yes. Each access specifier applies to members declared after it until another is encountered.
Example:
class A { private: int x; public: void get(); private: void show(); };

16. Analyze effect of modifying parameter using call by reference.
Changes in function affect the original variable.
Example:
void swap(int &a;,int &b;){int t=a;a=b;b=t;}
Original variables are changed.

17. Hierarchical Inheritance:
When multiple derived classes inherit from one base class.
Example:
class A{}; class B:public A{}; class C:public A{};
Diagram:
Base Derived1, Derived2

18. Single Inheritance:
One base class inherited by one derived

class. Example:
class A{public:void show();}; class B:public A{public:void display();}; Diagram: Base Derived

Part C – Essay Type Questions

19. a) Structure of C++ Program:
Header files, namespace, class, function declarations, main().
Example:

```
#include <iostream>

using namespace std;
int main(){ cout<<"Hello"; return 0; }
```

b) C++ Data Types:
- Basic: int, char, float, double
- Derived: array, pointer, reference, function
- User-defined: class, struct, union, enum

20. Swapping using call by value and call by reference:

Call by Value:

```
void swap(int a, int b){ int t=a; a=b; b=t; } // No change in originals
```

Call by Reference:

```
void swap(int &a;, int &b;){ int t=a; a=b; b=t; } // Changes originals
```

Difference: Value uses copies; Reference uses actual variables.

21. Define Base Class (Inheritance):
A base class is one whose members are inherited by another class.
Example:

```
class A{public:int x;}; class B:public A{}; Here, A is base class and B is derived class.
```